

YIFAN (IVAN) JING

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EDUCATION

University of Michigan, Ann Arbor, Michigan

Expected 2028

Winter

Bachelor of Science in Data Science

Research Interests: Reinforcement Learning, Language Models, Group Theory

RESEARCH EXPERIENCE

Terminal Bench-Science(<https://www.tbench.ai/news/tb-science-announcement>)

April 2026 - Current

Co-author and Contributor: Structure2Discover series tasks

- Contributing to Terminal-Bench Science, a pre-release scientific-workflow extension of Terminal-Bench, whose shipped benchmark has been adopted by frontier AI labs including Anthropic, OpenAI, and Google DeepMind.
- Developed the Structure2Discover task series to evaluate AI agents' ability to infer hidden structures from computational evidence, run terminal-based experiments, and produce verifiable scientific conclusions.

Hybrid Transformer Analysis in Induction Head Problem, University of Michigan

Sep 2025 - Dec 2025

Mentor: Dr. Samet Oymak

(details in this Github Repo <https://github.com/Ivann1242/Analysis-AR-problem>)

Researcher

- Studied generalization behavior of sequence models through the Associative Recall (AR) task, comparing Transformer, Mamba, and hybrid architectures.
- Proposed and evaluated hybrid architectures combining state-space models with attention mechanisms, demonstrating improved long-range generalization performance.
- Designed controlled experiments to analyze convergence and generalization dynamics across model families.

Contact Sensing Hexapod, BIRDS Lab, University of Michigan

Jan 2026 - Current

(I contribute simulation part of this work, <https://arxiv.org/abs/2603.09147>)

Mentor: Shai Revzen

Undergraduate Researcher

- Developed and conducted large-scale simulation experiments on Mujoco for a multi-legged robotic locomotion system, evaluating performance across varying leg counts (3 segments-64 segments) and terrain conditions.
- Implemented and tested modular control policies in simulation, validating stability and adaptability on irregular terrains.

TITO (Translation-Invariant Total Orders), University of Michigan

Sep 2025 - April 2026

(details in 2025 Fall:

<https://lsa.umich.edu/math/undergraduates/research-and-career-opportunities/research/LoGM/projects.html>)

Advisor: Dr. Grant Barkley

Researcher

- Developed a **new algorithmic approach** for Translation-Invariant Total Orders (TITO), formalizing order-comparability through inversion sets and proving its correctness.
- Translated the theoretical model into a **computable structure**, enabling efficient algorithms for sequence comparison and ranking, with potential applications in sorting and discrete optimization.
- Provide a python package for TITO weak order comparison, JOIN computation, Normalization, visualization

ThinkAct LLM Agent, University of Michigan

Sep 2025 - Dec 2025

Mentor: Dr. Samet Oymak

(details and research poster in this Github Repo <https://github.com/Ivann1242/ThinkAct>)

Co-contributor

- Improved reasoning performance of large language models by integrating self-reflection and self-consistency mechanisms within the ReAct framework.
- Designed evaluation pipelines to systematically analyze reasoning trajectories and performance gains on multi-step tasks.

Advisor: Dr. Aakash Gautam

Researcher

- Build a data democratization platform enabling non-technical users to access and interpret complex datasets.
- Using **javascript** to build the frontend page, implemented an online pipeline for automated preprocessing, visualization, and storytelling to simplify data-driven insights.
- Demonstrated applications on real datasets, bridging technical analysis with practical decision-making.

Customer Solution Recommendation, University of Pittsburgh, PA

March 2024 - August 2024

Director: Dr. Fatemeh Golshan

Research Assistant

- Develop a VIP customer recommender project which automated and personalized 80+ B2B customers and 36 product plans mix-and-match process, reduced 71% time cost of the average customer decision feedback loop.
- Utilized Javascript, HTML5, and CSS to conduct 62K+ big data cleaning, processing, and feature discovery via an interactive, real-time [website survey form](https://ivann1242.github.io/Webtool/).(https://ivann1242.github.io/Webtool/)

TEACHING EXPERIENCE

University of Pittsburgh, Pittsburgh, Pennsylvania

August 2024 - Dec 2024

Director: Dr. Paulo Brasko Ferreira

Undergraduate Teaching Assistant, Intermediate Programing

- Lectured undergraduate class of 125+ students; taught Java programming techniques in weekly office hours, requested recitations, and exam preparation sessions.
- Evaluated homework assignments, designed and graded exams and provided supporting online course materials.

RELEVANT EXPERIENCE/HONOR

Tartanhacks <https://github.com/danielryang/spaceoverflow>

- Led 4 undergraduate students from Brown Univ and GeTech to develop an innovative point-to-point warehouse space rental platform
- Win Top 5 Grant Over 300 submitted projects/700 participants in Tartanhacks 2026 hosted by CMU.

CAST-USA 33rd Innovation Summit

- Got Finalist and 1500\$ reward at this most significant Chinese Innovation Competition in the USA.
- Present CEEG AI Agent Business Plan in front of 300 audience.

She Innovations Hackathon

- Led 3 undergraduates from University of Pittsburgh to develop a 3D runner game in Unity and C# in a 2-day competition.

SKILLS

Languages: Python, C, C++, Java, Javascript, Mips

Tools: GitHub Actions, Jupyter Notebook, Colab, Google Cloud Platform, LaTeX